

3D Gas model

Maciej Stańczyk

5 kwietnia 2026

1 1-on-1 elastic collision

1.1 Detecting collision

The collision within the next step $t \in (0, \Delta t]$ is detected using the condition that at a time t the centers of mass of the two particles 1 and 2 are at the exact distance $r_1 + r_2$ where r_1 and r_2 are radii of the respective particles.

Position of the first particle at time t is:

$$\mathbf{X}_1(t) = \hat{\mathbf{X}}_1 + \mathbf{V}_1 t \quad (1)$$

where the hat denotes initial position (at time $t = 0$) and u , v and w are velocity components.

The position of the second particle can be derived similarly. The condition that the particles are touching can be now derived:

$$\left[\hat{\mathbf{X}}_1 - \hat{\mathbf{X}}_2 + t(\mathbf{V}_1 - \mathbf{V}_2) \right] \cdot \left[\hat{\mathbf{X}}_1 - \hat{\mathbf{X}}_2 + t(\mathbf{V}_1 - \mathbf{V}_2) \right] = (r_1 + r_2)^2$$

introducing notation:

$$\begin{aligned} \delta \hat{\mathbf{x}} &= \hat{\mathbf{X}}_1 - \hat{\mathbf{X}}_2, \\ \delta \mathbf{V} &= \mathbf{V}_1 - \mathbf{V}_2, \end{aligned}$$

one has:

$$[\delta \hat{\mathbf{x}} + t \delta \mathbf{V}] \cdot [\delta \hat{\mathbf{x}} + t \delta \mathbf{V}] = (r_1 + r_2)^2 \quad (2)$$

Solving for t one gets the quadratic:

$$t^2(\delta \mathbf{V} \cdot \delta \mathbf{V}) + t(2\delta \mathbf{V} \cdot \delta \hat{\mathbf{X}}) + (\delta \hat{\mathbf{X}} \cdot \delta \hat{\mathbf{X}}) - (r_1 + r_2)^2 = 0 \quad (3)$$

This is solved in a standard method. If a solution exists for $t \in (0, \Delta t]$, the smaller one is selected (in case there are two). This time t_c determines the instant of the collision. Equation 1 can be used to determine collision location.

1.2 Collision energy and momentum transfer

The collision momentum and energy transfer for a fully elastic collision is described by conditions of momentum and energy conservation. Before formulating those, we introduce new inertial coordinate system $\bar{X}$ such as:

$$\bar{\mathbf{X}} = \mathbf{X} - \mathbf{V}_2(t - t_c) \quad (4)$$

where velocity $\mathbf{V}_2$ of the second particle is taken before the collision. By differentiating these equations for the second particle we see that it is stationary prior to the collision in this new coordinate system.

Position of the first particle in this system is, after substituting equation 1:

$$\bar{\mathbf{X}}_1 = \hat{\mathbf{X}}_1 + t\delta\mathbf{V} + t_c\mathbf{V}_2$$

and its velocity is obviously:

$$\bar{\mathbf{V}}_1 = \delta\mathbf{V}$$

The versor $\bar{\mathbf{p}}$ is introduced, defining the direction from the center of particle 1 to center of particle 2 at the moment of collision:

$$\bar{\mathbf{p}} = -\frac{1}{r_1 + r_2}(\delta\hat{\mathbf{X}} + t_c\delta\mathbf{V})$$

Now we define the conservation laws in the inertial coordinate system $\bar{X}$:

$$m_1\bar{\mathbf{V}}_1 = m_1\bar{\mathbf{V}}_{1c} + m_2\bar{\mathbf{V}}_{2c} \quad (5)$$

$$m_1(\bar{\mathbf{V}}_1 \cdot \bar{\mathbf{V}}_1) = m_1(\bar{\mathbf{V}}_{1c} \cdot \bar{\mathbf{V}}_{1c}) + m_2k_c^2 \quad (6)$$

$$\bar{\mathbf{V}}_{2c} = k_c\bar{\mathbf{p}} \quad (7)$$

where the added index c denotes velocities after collision and k_c indicates the speed of the second particle after collision.

The first equation is momentum conservation, the second expresses energy conservation, the last simply captures the fact, that after the collision the second particle will depart along the direction $\bar{\mathbf{p}}$. These need to be solved for $\mathbf{V}_{1c}$ and $\mathbf{V}_{2c}$ (k_c is also unknown).

Introducing notation $\varphi = \frac{m_2}{m_1}$, using that $k_c \neq 0$, and noting that, by definition of versor $\bar{\mathbf{p}} \cdot \bar{\mathbf{p}} = 1$:

$$\begin{aligned} k_c &= \frac{2(\bar{\mathbf{v}}_1 \cdot \bar{\mathbf{p}})}{\varphi + 1} \\ \bar{\mathbf{v}}_{1c} &= \bar{\mathbf{v}}_1 - \varphi k_c \bar{\mathbf{p}} \\ \bar{\mathbf{v}}_{2c} &= k_c \bar{\mathbf{p}} \end{aligned}$$